

POLITECNICO
MILANO 1863

Contiki-NG Evaluation Lab

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A Network Data Structure

- You are to implement a network data structure that allows modifications from remote clients
- The data structure offers three operations to remote clients
  - **`lock`**: locks the data structure for possible write operations by remote clients
  - **`write(value)`**: assigns a new value to the data structure, provided by the remote clients
  - **`read`**: returns the current value of the data structure to remote clients

Locks and Writes

- A **`write`** operation is possible if and only if the data structure is **`lock`**-ed by the same remote client issuing the write
  - **`write`** operations without a **`lock`** should be refused and an **error code** returned to the remote client
  - A **`lock`** is lifted on the data structure after either completing a **`write`** or after a 5 seconds timeout
- If the data structure is **`lock`**-ed and another **`lock`** request comes in, nothing happens
- Instead, **`read`** operations are possible regardless of the state of the lock

Suggestions and Tips

- Start from **`udp-server`** and **`udp-client`** examples
- Use the server to store the data structure and to offer operations to remote clients
- Test with two different clients issuing a predetermined test sequence
- The data structure may be a single **`int`** value

Suggestions and Tips

- Use COOJA and COOJA motes
- Useful functions
  - **`#include "net/routing/rpl-lite/rpl.h"`**
  - **`rpl_parent_get_ipaddr (curr_instance.dag.preferred_parent)`**
    returns the IPv6 address of the current parent
  - **`uip_ipaddr_cmp(*addr1,*addr2)`**
    compares two IPv6 addresses
  - **`uip_ipaddr_copy(*dest,*source)`**
    copies IPv6 **`source`** address into **`dest`**

Rules

- Complete the README.md file with
  - Your group identifier
    - From the group registration document
  - Name of each group member
  - A 200-word (max) description of your solution
    - What piece of code does what when…
- Create and submit a single zip file with
  - The README.md
  - All necessary source files and Makefile(s) in no specific subdirectory
  - A `.csc` file with a test topology including the server and two clients
  - Remove everything that is not the project you want to submit
    - Binary files…
  - Name of the file: contiking-groupXX.zip
    - XX is the group identifier from the group registration document
  - Submit by the user corresponding to the contact email specified in the group registration document